# Unit 13, Lesson 8: Classifying Quadrilaterals – Part 1

## Benchmark

MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals.

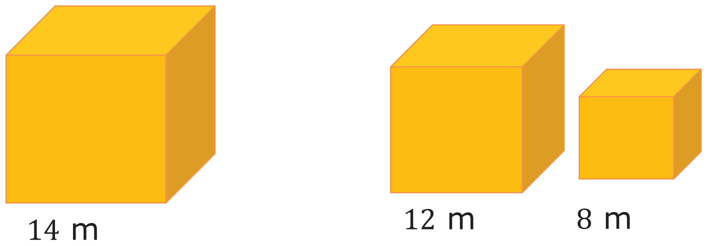
## Learning Targets

- ☐ I can use coordinate geometry to classify quadrilaterals based on their properties in a mathematical or real-world context.
- ☐ I can draw a quadrilateral on a coordinate plane using given definitions, properties, or theorems.

## 13.8.1 Warm-Up: Would You Rather?

Gold is worth a significant amount of money. Its value changes with the market. For example, on January 28, 2021 the value of $1$ ounce of gold was nearly $\$2000$.

1. Would you rather have one cube made of solid gold measuring $14$ meters (m) on each side, or two cubes made of solid gold, one measuring $12$ m on each side and one measuring $8$ m on each side?




2. Explain your choice.

## 13.8.2 Exploration: Classifying Quadrilaterals Using Properties

1. Complete the table by filling in the blanks using the word bank. Words may be used more than once, or they may not be used.

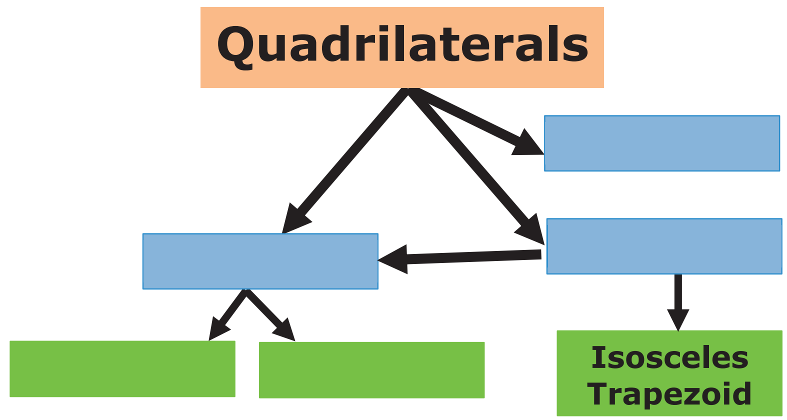
| Word Bank | |
|---|---|
| bisect(s) | congruent |
| parallel | perpendicular |

| Classification | Property |
|---|---|
| Parallelogram | Opposite sides are _______________ and ____________________. |
| Parallelogram | Diagonals ______________ each other. |
| Rectangle | Diagonals are _____________________. |
| Rhombus | Diagonals are _____________________. |
| Square | Diagonals are ___________________ and ______________________. |
| Kite | Exactly one diagonal _______________ the other, and diagonals are ________________________. |
| Isosceles Trapezoid | Diagonals are ___________________ and at least one pair of sides are ________________. |

2. Discuss with your partner if knowing two consecutive sides are congruent is sufficient to classify a quadrilateral. Summarize your discussion and justify your decision.

3. Discuss with your partner if knowing a pair of opposite sides are congruent is sufficient to classify the quadrilateral. Summarize your discussion and justify your decision.

4. Complete the diagram using the different types of quadrilaterals. Each blue box is a subcategory of quadrilaterals. Each green box is a subcategory of the blue box that it is connected to.



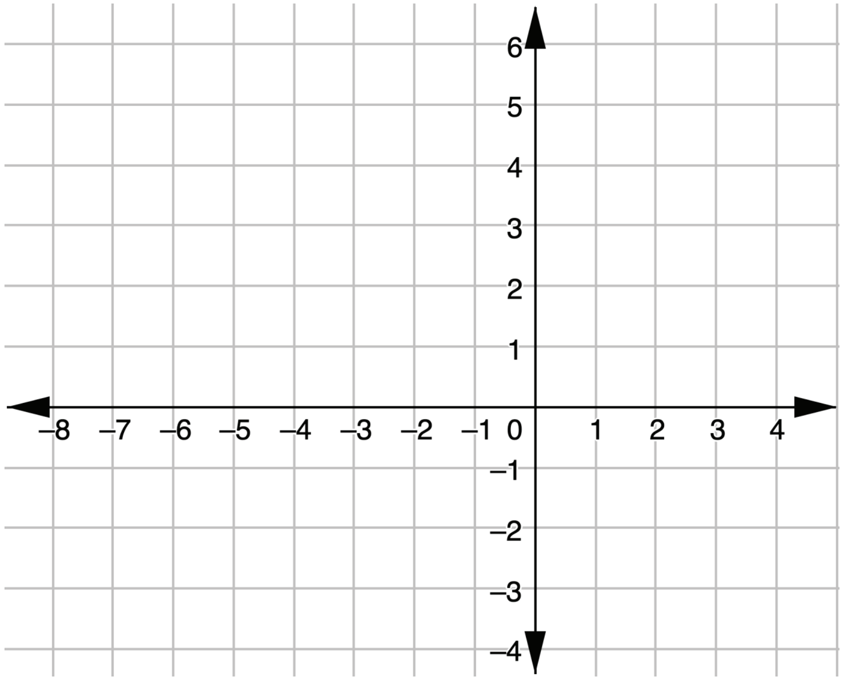

5. Which quadrilateral was not included in the diagram?

6. Draw a box and any necessary arrows to include the missing quadrilateral in the diagram.

## 13.8.3 Exploration: Classifying Quadrilaterals

1. $RMWB$ has vertices at $R(-6,2)$, $M(-4,5)$, $W(-1,3)$, and $B(-3,0)$.

   a. Graph $RMWB$ on the coordinate plane.




   b. Complete the table.

| Line Segment | Length |
|---|---|
| $\overline{RM}$ | |
| $\overline{MW}$ | |
| $\overline{WB}$ | |
| $\overline{BR}$ | |

   c. What do you notice about the side lengths of $RMWB$?

   d. Select all of the classifications that apply to $RMWB$.

   - ☐ Isosceles trapezoid
   - ☐ Kite
   - ☐ Parallelogram
   - ☐ Rectangle
   - ☐ Rhombus
   - ☐ Square
   - ☐ Trapezoid

   e. Complete the table.

| Diagonal Line Segment | Length |
|---|---|
| $\overline{RW}$ | |
| $\overline{MB}$ | |

   f. What do you notice about the diagonal lengths of $RMWB$?

g. Complete the statement.

Since $RMWB$ ○ has / ○ does not have congruent diagonals, $RMWB$ is classified as a(n)

____________________.

2. $NTCV$ has vertices at $N(2,3)$, $T(3,5)$, $C(6,6)$, and $V(4,2)$.

a. Complete the table.

| **Line Segment** | **Length** |
|---|---|
| $\overline{NT}$ | |
| $\overline{TC}$ | |
| $\overline{CV}$ | |
| $\overline{VN}$ | |

b. What do you notice about the side lengths of $NTCV$?

c. Complete the table.

| **Line Segment** | **Slope** |
|---|---|
| $\overline{NT}$ | |
| $\overline{TC}$ | |
| $\overline{CV}$ | |
| $\overline{VN}$ | |

d. What do you notice about the slopes of $NTCV$?

e. Select all of the classifications that apply to $NTCV$.

☐ Isosceles trapezoid
☐ Kite
☐ Parallelogram
☐ Rectangle
☐ Rhombus
☐ Square
☐ Trapezoid

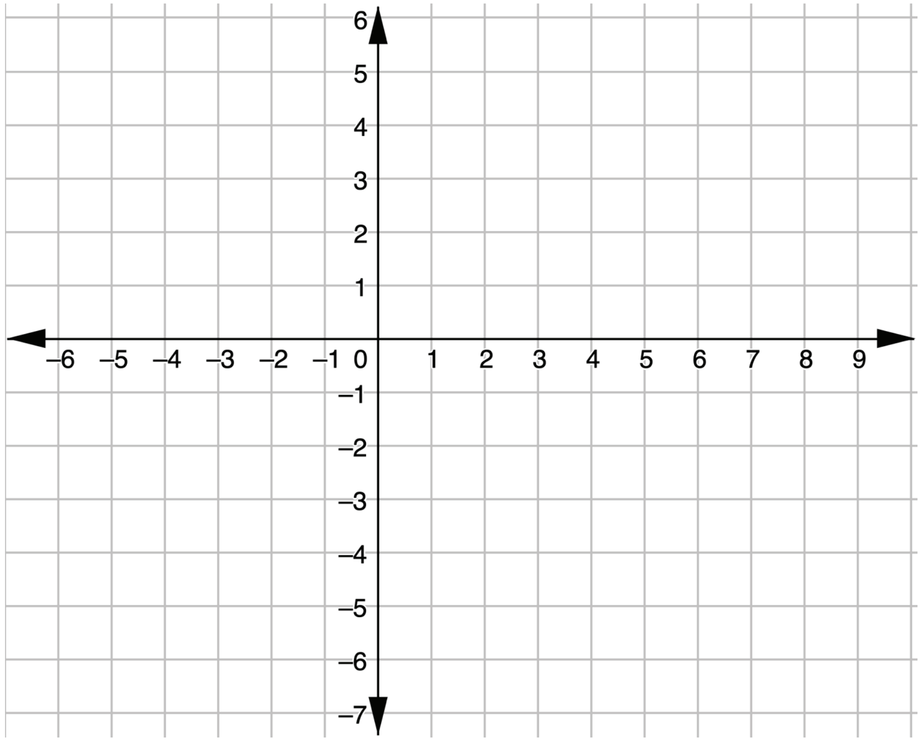
## 13.8.4 Guided Practice: Classifying Quadrilaterals

1. $GSPD$ has vertices at $G(-2,-3)$, $S(3,-2)$, $P(5,-4)$, and $D(0,-5)$.

   Justify that $GSPD$ is a parallelogram but not a rectangle. Show your work.

2. $HMTK$ has vertices at $H(-2,4)$, $M(-2,0)$, $T(-6,-1)$, and $K(-6,5)$. Classify $HMTK$ and justify your classification using mathematics.

3. Use the coordinate grid to complete the following.
   a. Draw a quadrilateral for which the side lengths are all equal, but the diagonals are not congruent.



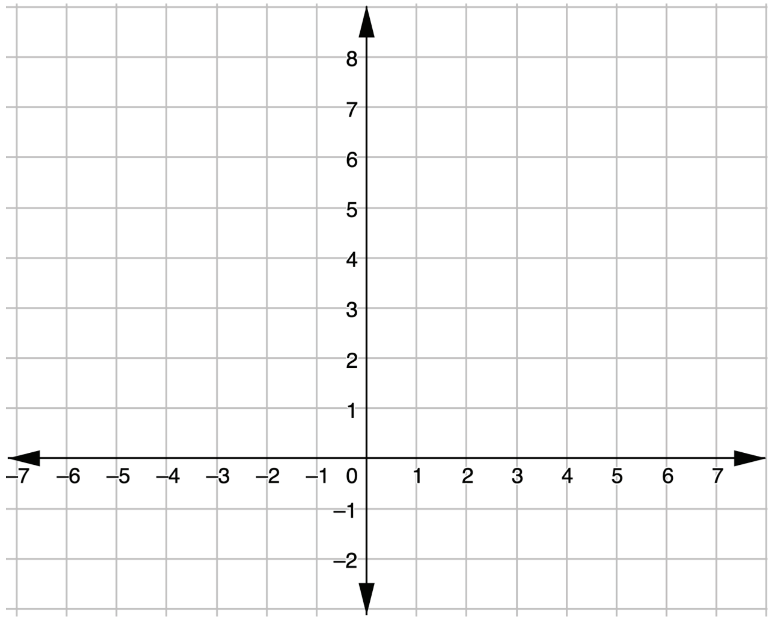

   b. Complete the statement.

   This quadrilateral is classified as a ________________.

4. Discuss with your partner how another classmate could use the coordinates of your quadrilateral to verify that your classification is correct. Summarize your discussion.

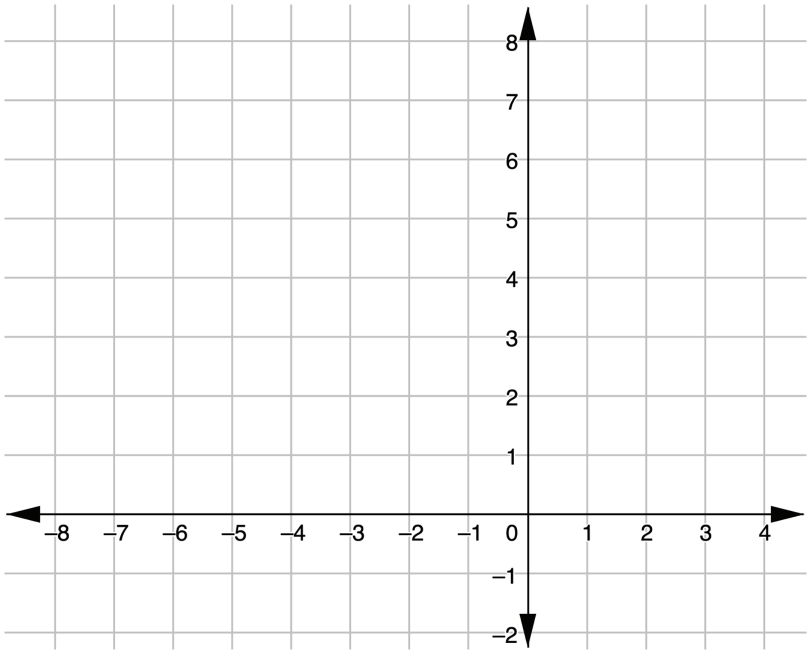
## 13.8.5 Your Turn!

1. $KHCY$ has vertices at $K(1,6)$, $H(3,4)$, $C(1,1)$, and $Y(-1,4)$.




$KHCY$ is classified as a _______________ because …

2. Draw a quadrilateral with one set of parallel sides and vertices at point $(-4,7)$ and $(-3,5)$.

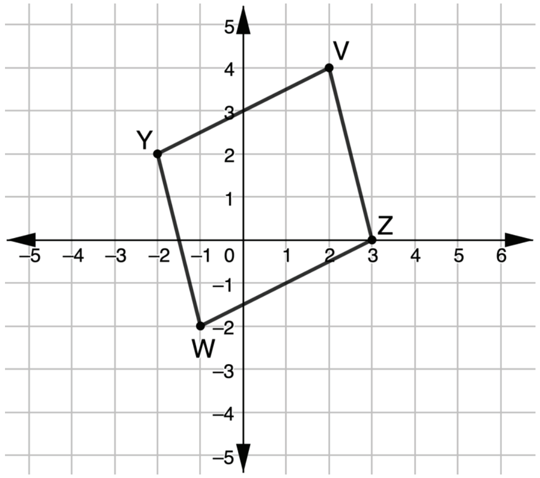


This quadrilateral is classified as a _______________ because …

## 13.8.6 Wrap-Up: Error Analysis

1. $YVZW$ has vertices at $Y(-2,2)$, $V(2,4)$, $Z(3,0)$, and $W(-1,-2)$.

   Aryana graphed $YVZW$ and classified the quadrilateral as a rhombus.




   a. Do you agree with Aryana?

   b. Justify your answer using properties and showing your work.